

SEWARD, AK, USA

WMO: 702770

Lat: 60.128N

Lon: 149.417W

Elev: 22

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26438

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.4	(c) 8.1	(d) -10.9	(e) 3.1	(f) 8.3	(g) -7.6	(h) 3.7	(i) 10.3	(j) 33.9	(k) 12.9	(l) 31.4	(m) 15.2	(n) 19.7	(o) 20	(p) 0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	70.8	57.2	66.6	56.1	63.7	55.6	59.9	67.1	58.3	64.3	57.0	62.2	10.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.6	68.3	61.6	55.2	64.9	59.8	54.4	63.0	58.8	26.2	66.7	25.2	64.4	24.3	62.1	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.9	21.6	DB	1.6	79.4	6.2	5.2	-2.9	83.1	-6.5	86.1	-10.0	89.0	-14.5	92.8
			WB	0.7	62.2	6.0	3.8	-3.6	64.9	-7.1	67.1	-10.5	69.3	-14.9	72.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	26.4	28.7	31.0	38.2	46.0	52.2	56.0	55.7	49.7	40.4	31.6	27.6
	DBStd	12.59	10.02	8.47	7.19	4.83	5.13	4.68	4.20	3.96	4.06	5.36	7.62	8.19
	HDD50	4040	731	597	589	354	148	22	1	1	53	299	551	694
	HDD65	9002	1196	1017	1054	803	589	386	285	289	460	763	1001	1159
	CDD50	523	0	0	0	1	24	88	186	180	43	1	0	0
	CDD65	9	0	0	0	0	0	2	5	2	0	0	0	0
	CDH74	60	0	0	0	0	2	12	32	14	0	0	0	0
	CDH80	8	0	0	0	0	0	2	4	2	0	0	0	0
Wind	WSAvg	8.5	10.9	9.9	10.1	7.9	7.4	6.5	5.8	6.2	7.1	8.6	10.8	11.1
Precipitation	PrecAvg	65.4	6.3	5.4	4.0	3.8	4.3	2.5	2.2	5.3	9.9	10.6	6.3	7.4
	PrecMax	96.4	25.4	13.3	12.3	9.7	15.9	8.8	6.0	14.1	26.1	24.0	25.2	19.7
	PrecMin	36.4	0.7	0.1	0.2	0.3	0.4	0.6	0.2	0.9	3.1	2.1	0.2	0.9
	PrecStd	15.3	5.8	3.5	2.9	2.9	3.3	1.7	1.5	3.5	5.3	5.7	5.6	5.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	43.9	48.8	59.4	69.9	74.5	79.1	75.1	65.8	55.0	48.0	44.8
		MCWB	39.4	38.8	41.2	45.9	52.2	58.6	61.4	60.0	53.5	46.6	43.6	41.1
	2%	DB	42.4	41.4	44.5	52.3	63.1	68.2	70.4	69.9	61.4	51.9	45.3	42.4
		MCWB	38.8	37.6	38.8	42.5	49.4	55.7	58.3	57.1	51.7	45.3	42.6	39.4
	5%	DB	40.0	39.4	41.9	48.6	58.4	63.4	66.0	65.8	58.4	49.7	43.4	40.1
		MCWB	37.4	36.2	37.3	41.5	48.3	54.7	57.6	56.1	51.3	44.9	41.0	37.7
	10%	DB	37.9	37.6	39.4	46.0	54.7	60.5	63.2	63.1	55.8	47.9	41.3	37.8
		MCWB	36.0	35.1	35.7	40.1	47.0	53.4	56.9	56.2	50.5	44.2	39.1	36.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.7	40.3	43.2	48.1	53.6	60.4	62.2	61.9	55.8	50.3	45.6	42.3
		MCDB	44.1	42.4	46.6	57.9	65.2	70.9	74.5	70.5	60.6	52.4	46.3	43.8
	2%	WB	39.5	38.4	40.5	44.3	51.1	57.3	60.1	59.7	54.2	47.9	43.3	39.7
		MCDB	41.5	40.6	43.3	49.7	59.3	64.6	67.9	66.2	58.0	49.5	44.7	41.7
	5%	WB	37.9	37.1	38.2	42.4	49.5	55.6	58.7	58.2	53.1	46.3	41.6	38.0
		MCDB	39.6	39.0	40.7	47.0	56.2	61.9	64.7	63.8	56.4	48.0	43.4	39.7
	10%	WB	36.3	35.8	36.6	41.1	47.9	53.9	57.3	56.8	51.9	45.1	39.3	36.3
		MCDB	37.7	37.6	38.9	45.3	53.8	59.0	62.5	61.5	55.0	47.1	41.0	37.8
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	11.0	12.2	13.0	12.1	10.8	11.6	11.2	10.2	8.1	8.5
		MCDBR	9.6	10.1	12.7	17.4	20.6	19.9	19.1	18.7	15.4	12.2	9.2	9.8
		MCWBR	6.9	7.1	8.6	10.4	10.7	10.5	9.3	9.1	8.7	7.4	6.9	7.3
	5% WB	MCDBR	9.1	8.9	11.3	14.8	16.5	16.7	15.7	15.3	12.2	9.1	8.6	9.3
		MCWBR	6.9	6.7	8.2	9.7	9.6	9.4	8.6	8.9	7.8	6.7	6.9	7.3
Clear-Sky Solar Irradiance	taub	0.260	0.315	0.329	0.370	0.389	0.387	0.390	0.378	0.337	0.317	0.266	0.257	
	taud	2.081	2.097	2.108	2.047	2.100	2.252	2.303	2.338	2.434	2.269	2.070	1.942	
	Ebn at Noon	156	207	248	259	263	267	263	257	250	209	147	98	
	Edh at Noon	14	25	34	43	44	39	36	33	25	21	14	10	
All-Sky Solar Radiation	RadAvg	99	304	771	1253	1628	1661	1486	1175	784	374	142	56	
	RadStd	13	53	98	143	136	90	136	137	118	42	24	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.56	+7.04	+6.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+1.22	+3.87	+3.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page